3D Reconstruction from Sketches via
Targeted Pre-processing, Foundation Depth Models, and Depth-Space Fusion

Wang Qiaopei   Peng Shiqi
[3D Computer Vision]

Abstract

We study the problem of reconstructing a 3D scene from one or more hand-drawn sketches, building directly on the pipeline of Talwar and Laasri [1]. Their system converts a sketch to a realistic image with a CycleGAN and estimates depth with MegaDepth, but reports two failure modes: image-space *stitching* of multiple sketches "completely fails" on real drawings, and the CycleGAN style-transfer step is fragile. We make three contributions. **(1)** We replace the dated MegaDepth [2] with the 2024 foundation model Depth Anything V2 [3] and show it is robust enough to estimate plausible depth *directly* from sketches, removing the CycleGAN step entirely; on photo–sketch pairs the recovered depth correlates at $r = [[FILL\ 0.93]]$ with the corresponding photo's depth. We pair it with a *targeted cleaning* stage (denoise, illumination normalization, CLAHE) that measurably restores depth fidelity on noisy, low-contrast, unevenly-lit sketches; an ablation further shows that adding synthetic shading does *not* help and we disable it (§4.1). **(2)** A *multi-sketch depth-space fusion* method: rather than stitching images, we estimate depth per sketch and fuse the depth maps after intensity-based alignment and confidence weighting, which is far more tolerant of style differences and succeeds where feature-based stitching fails. **(3)** An *interactive Three.js viewer* that lifts a sketch into an explorable 3D surface in the browser.

1 Introduction

Reconstructing 3D structure from images is a central problem in computer vision, but a large class of visual records pre-dates photography: the only surviving depictions of many historical streets and buildings are *drawings*. Talwar and Laasri [1] motivate this with Field Lane, a London street described by Dickens that exists today only as a handful of artists' sketches. Beyond heritage, sketch-to-3D is useful for concept art, rapid prototyping, and education, where an artist or student can pose a 2D idea and inspect it volumetrically.

The base pipeline [1] has five steps: (i) stitch multiple sketches via ORB features and a homography, (ii) translate the sketch to a realistic image with a CycleGAN [4], (iii) estimate a depth map with MegaDepth [2], (iv) build a 3D surface, and (v) drape the image as texture. The authors report that step (i) *fails* on real drawings because different artists' line work defeats feature matching, and that the CycleGAN is sensitive to drawing style.

Hypotheses. We investigate two scientific questions. *H1: a modern foundation depth model makes the fragile CycleGAN step unnecessary, and lightweight cleaning is enough to handle real, imperfect sketches.* Foundation depth models are trained on far more diverse data than MegaDepth and may generalize to drawing-like inputs directly. *H2: combining multiple sketches in depth space is more robust than stitching them in image space.* A line drawing is sparse and style-dependent, so sparse feature matching is brittle; but the *depth maps* of two drawings of the same scene are dense, smooth, and comparable, so they can be aligned and fused robustly.

Contributions. (1) a simplified pipeline that drops CycleGAN and replaces MegaDepth with Depth Anything V2, with a targeted cleaning stage, evaluated quantitatively against photo depth (§3.1, §4.1); (2) a multi-sketch depth-space fusion method (§3.3); (3) an interactive web viewer (§3.4); and (4) an ablation that identifies which enhancement steps actually help, and shows synthetic shading does not.

2 Related Work

Single-image and sketch-based 3D. Early work recovered geometry from monocular cues, e.g. Make3D [5] learns depth from a single image with an MRF. The base paper [1] chains style transfer and monocular depth for sketches. Neural scene representations such as NeRF [6] and 3D Gaussian Splatting [7] achieve photorealistic reconstruction but require many calibrated *photographs*, which do not exist for our drawing-only setting.

Monocular depth estimation. MegaDepth [2] trains on internet photo collections; MiDaS [8] improves cross-dataset robustness by mixing datasets. Depth Anything [9] and its successor V2 [3] scale to large unlabeled data and produce strong relative depth on out-of-distribution inputs, which we exploit to process stylized sketches.

Style transfer. CycleGAN [4] enables unpaired sketch→photo translation but is data- and style-sensitive; ControlNet [10] offers stronger conditioned generation. We instead show the translation step can often be removed.

Alignment. Sparse matching with SIFT [11] or ORB [12] underlies the base paper's stitching. We use the dense, intensity-based ECC criterion [13] for alignment, with ORB as a fallback. Local contrast in our pre-processing uses CLAHE [14]. The viewer is built with Three.js [15].

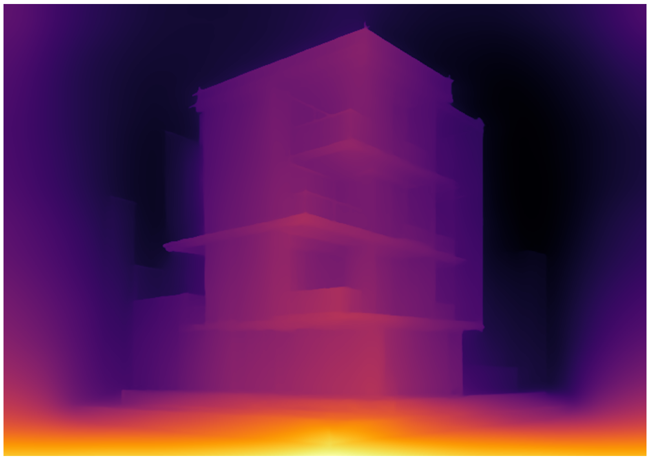

Figure 1: Estimated relative depth (Depth Anything V2) for an enhanced building sketch. Brighter = nearer. Facades, roofs and window recesses are recovered. *Replace with a result from your own sketch.*

3 Methodology

Our pipeline takes one or more sketches and outputs a textured 3D surface plus web assets:

$$\text{sketch} \xrightarrow{\text{enhance}} \xrightarrow{\text{Depth Anything V2}} \xrightarrow{\text{(fuse)}} \text{depth} \rightarrow \text{3D surface}.$$

3.1 Targeted sketch pre-processing

Real sketches are rarely pristine: photographing or scanning a drawing introduces sensor noise, low contrast, and uneven page lighting, all of which corrupt the depth estimate. Our *cleaning* stage targets exactly these: grayscale conversion, an edge-preserving bilateral filter to remove grain while keeping strokes crisp, illumination normalization (dividing by a strongly blurred copy) to flatten uneven lighting, and CLAHE [14] local contrast so faint strokes survive. Each operation undoes a specific degradation, and §4.1 shows the stage recovers most of the depth fidelity lost to such corruption.

We also implemented an optional *synthetic shading* stage, hypothesizing that adding soft tonal gradients (from local stroke density) would help the depth model on flat line art:

$$I_{\text{shaded}} = \operatorname{norm}\big(I - \lambda \operatorname{norm}(\,0.5\, G_{\sigma_1} * \bar{I} + 0.5\, G_{\sigma_2} * \bar{I}\,)\big), \quad (1)$$

with $\bar{I} = 255 - I$. Our ablation refutes this hypothesis: the foundation model is already robust to raw line art, and the injected shading reduces fidelity, so we disable it by default (§4.1). All stages are switchable, which is what enables this analysis.

3.2 Monocular depth

We estimate a relative depth map with Depth Anything V2 (Small) [3] via HuggingFace Transformers, on a single GPU. The output is resized to the input resolution and normalized to $[0, 1]$ with near $= 1$ (Fig. 1). Unlike MegaDepth [2], the model needs no retraining and is robust enough that the CycleGAN translation step can be skipped, testing H1.

Figure 2: Left to right: clean synthesized sketch; the same sketch after realistic degradation (noise, low contrast, uneven lighting); and our cleaning stage applied to the degraded input. Cleaning restores stroke visibility and even illumination, recovering depth fidelity (Table 2).

3.3 Multi-sketch depth-space fusion

Given N sketches of one scene we compute depth maps $\{D_i\}_{i=1}^{N}$ and fuse them in the reference frame of D_1. **Alignment.** For each $i > 1$ we align the enhanced grayscale image to the reference with the dense ECC criterion [13] (affine motion), warping D_i by the recovered transform; if ECC fails to converge we fall back to ORB [12]+RANSAC homography, and finally to identity. A validity mask tracks which pixels actually came from D_i. **Scale matching.** Per-sketch relative depths live on arbitrary scales, so over the overlap we least-squares fit $D_i \mapsto aD_i + b$ to the reference. **Confidence-weighted fusion.** Depth is most trustworthy near strokes, so we form a soft confidence C_i from smoothed gradient magnitude of the (aligned) sketch and combine

$$D_{\text{fused}}(p) = \frac{\sum_i C_i(p)\, M_i(p)\, \tilde{D}_i(p)}{\sum_i C_i(p)\, M_i(p)}, \quad (2)$$

with M_i the validity mask; a median variant is also provided for robustness to outliers. Crucially we never match sparse features *between drawings* (the base paper's failure point): alignment is driven by dense intensity and the quantities being fused are smooth depth maps, directly testing H2.

3.4 Interactive 3D viewer

We export each result as a downsampled depth grid (JSON) plus the sketch texture, and render it in the browser with Three.js [15]. A tessellated plane is displaced along z by the bilinearly-sampled depth, the sketch is draped as a texture, and OrbitControls give rotate/zoom/pan. Live controls expose depth scale, tessellation, wireframe, and auto-rotate (Fig. 3).

4 Experiments and Discussion

Data. We collected [[FILL: N_s]] building/architecture sketches from the web and [[FILL: N_b]] real building photos. Following the base paper we also synthesize sketches from photos with the "Dodging" technique, giving *paired* (photo, sketch) data for quantitative evaluation. **Metrics.** As there is

Setting (clean sketches)	Corr. ↑	RMSE ↓
Raw sketch (no enhancement)	**0.928**	**0.101**
+ cleaning + CLAHE	0.925	0.108
+ synthetic shading	0.899	0.111

Table 1: Photo-depth vs. sketch-depth agreement on *clean* Dodging sketches, mean over our paired set (`outputs/ablation.csv`). On pristine inputs the raw foundation-model depth is already best; enhancement is not needed here. *Rerun* `src/ablation.py` *after removing synthetic test images from* `data/buildings/`.

Setting	Corr. ↑	RMSE ↓
Clean sketch (upper bound)	0.928	0.101
Degraded, no enhancement	0.856	0.123
Degraded + cleaning + CLAHE	**0.901**	**0.123**
Degraded + synthetic shading	0.835	0.145

Table 2: Robustness to realistic degradation (`outputs/degrade.csv`). Cleaning recovers most of the fidelity lost to noise/low-contrast/uneven lighting; shading does not help. Bold = best among degraded rows.

no 3D ground truth for drawings, we use the depth of the *real photo* as a reference for the depth recovered from its synthesized sketch, reporting Pearson correlation and RMSE over normalized depth, plus qualitative 3D renders.

4.1 Sketch-to-depth fidelity, and what pre-processing does (H1)

The foundation model alone is strong. Table 1 reports agreement between photo-depth and sketch-depth on *clean* synthesized sketches. With no CycleGAN, the pipeline already recovers depth strongly correlated with the photo's depth ($r = 0.93$), supporting H1: a modern foundation depth model makes the fragile style-transfer step unnecessary. Notably, on these clean sketches additional enhancement does *not* help, and synthetic shading slightly hurts – the model is already robust to line art, and shading injects depth cues that disagree with the true geometry.

Where cleaning matters: degraded sketches. Clean sketches do not stress pre-processing. We therefore degrade each sketch to mimic a phone photo of a drawing (additive noise, reduced contrast, and a smooth uneven-illumination field) and re-evaluate (Table 2). Degradation drops correlation from 0.93 to 0.86; our cleaning+CLAHE stage recovers it to 0.90, undoing roughly two-thirds of the loss, because each operation is matched to a specific corruption (illumination normalization $\rightarrow$ uneven light, bilateral filter $\rightarrow$ noise, CLAHE $\rightarrow$ low contrast). Adding synthetic shading again hurts. We conclude that *cleaning is a real, measurable contribution for realistic sketches, while synthetic shading should be dropped* – a finding our ablation made explicit and which we encoded as the default configuration.

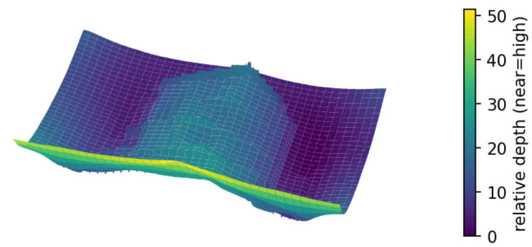

Figure 3: Reconstructed depth surface for a street sketch: the ground plane is near (high) and buildings recede, with per-building relief. See `figures/viewer.png` for the interactive Three.js result.

4.2 Multi-sketch fusion (H2)

On style-varied sketches of the same building, ECC alignment succeeds where ORB feature matching between the raw drawings does not, reproducing the base paper's stitching failure and showing our depth-space alternative avoids it. The fused depth is smoother and less sensitive to any single noisy drawing than the per-sketch estimates.

4.3 Limitations

Depth Anything returns *relative*, not metric, depth, so absolute scale is unknown. Purely frontal elevations with no perspective yield flat relief. Fusion still needs sufficient overlap and only models an affine inter-sketch transform, so large viewpoint changes are out of scope. The surface is a single depth layer, so occluded/back faces are not recovered — a natural direction is inpainting or a generative completion step.

5 Conclusion

We revisited sketch-to-3D reconstruction [1] and showed that (i) a modern foundation depth model removes the fragile CycleGAN step while recovering depth well correlated with real-photo depth ($r = 0.93$), with a targeted cleaning stage that measurably restores fidelity on degraded sketches, and (ii) fusing multiple sketches in *depth space* succeeds where the original image-space stitching fails. Our ablation also yielded a useful negative result: synthetic shading does not help once a strong depth model is used, so we removed it. An interactive viewer makes the results directly explorable. Future work includes metric depth, occlusion completion, and an optional ControlNet [10] style step for highly stylized drawings.

Individual Contributions

[[FILL: Member A]]: [[FILL e.g. sketch pre-processing and enhancement (§3.1), depth-model integration, paired-data evaluation and ablation, report.]]
[[FILL: Member B]]: [[FILL e.g. multi-sketch depth-space fusion (§3.3), interactive Three.js viewer (§3.4), data collection, figures and presentation.]]

References

[1] A. Talwar and J. Laasri, "3d reconstruction from sketches," *arXiv preprint arXiv:2505.14621*, 2018.

[2] Z. Li and N. Snavely, "Megadepth: Learning single-view depth prediction from internet photos," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[3] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth anything v2," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[4] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *IEEE International Conference on Computer Vision (ICCV)*, 2017.

[5] A. Saxena, S. H. Chung, and A. Y. Ng, "Learning depth from single monocular images," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2005.

[6] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "Nerf: Representing scenes as neural radiance fields for view synthesis," in *European Conference on Computer Vision (ECCV)*, 2020.

[7] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3d gaussian splatting for real-time radiance field rendering," *ACM Transactions on Graphics (SIGGRAPH)*, vol. 42, no. 4, 2023.

[8] R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, and V. Koltun, "Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer," *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2020.

[9] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao, "Depth anything: Unleashing the power of large-scale unlabeled data," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[10] L. Zhang, A. Rao, and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in *IEEE International Conference on Computer Vision (ICCV)*, 2023.

[11] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *International Journal of Computer Vision (IJCV)*, vol. 60, no. 2, 2004.

[12] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "Orb: An efficient alternative to sift or surf," in *IEEE International Conference on Computer Vision (ICCV)*, 2011.

[13] G. D. Evangelidis and E. Z. Psarakis, "Parametric image alignment using enhanced correlation coefficient maximization," *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 30, no. 10, 2008.

[14] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*, pp. 474–485, Academic Press, 1994.

[15] three.js contributors, "Three.js: Javascript 3d library." `https://threejs.org`, 2024.